Assignment #1

1. Describe the standard algorithm for finding the binary representation of a positive decimal integer.
   a. in English.
   b. in pseudocode.

2. Consider the following algorithm for finding the distance between the two closest elements in an array of numbers. Make as many improvements as you can in this algorithmic solution to the problem. If you need to, you may change the algorithm altogether; if not, improve the pseudecode given.

ALGORITHM *MinDistance*$(A[0..n-1])$

```
//Input: Array A[0..n – 1] of numbers
//Output: Minimum distance between two of its elements
dmin ← ∞
    for i ← 0 to n – 1 do
        for j ← 0 to n – 1 do
            if i ≠ j and |A[i] – A[j]| < dmin
                dmin ← |A[i] – A[j]|
return dmin
```